

AOPs as Value-Adders Beyond Standalone Mechanistic NAMs: A Cardiovascular Perspective

The Core Problem with Standalone NAMs

Mechanistic NAMs — hERG patch clamp, calcium imaging, mitochondrial membrane potential assays, contractility readouts in hiPSC-CMs — are excellent at detecting perturbation of a specific biological target. The limitation is that a positive signal in isolation answers only the question "does this compound touch this target?" It does not answer the regulatory question: "does this perturbation lead to an adverse cardiac outcome, through what chain of events, and under what conditions?"

This is where AOPs provide structural added value. An AOP is not another assay. It is a **causal biological narrative** — a formally curated, evidence-weighted chain from a Molecular Initiating Event (MIE) through Key Events (KEs) to an Adverse Outcome (AO) — that tells you *which* assays to run, *in what order*, and *what the positive result means* in terms of human health risk.

Concrete AOP Examples from AOP-Wiki for Cardiovascular Toxicity

The AOP-Wiki currently hosts a rich cluster of cardiac AOPs converging on heart failure (KE1535) and decreased left ventricular function (KE2215) as adverse outcomes. Three mechanistically distinct pathways illustrate the spectrum:

1. AOP261 — L-Type Calcium Channel Blockade → Heart Failure via Contractility Decrease

Chain: KE1529 (Blockade, L-Type Ca^{2+} channels, *Molecular*) → KE1530 (Decrease, Ca^{2+} currents, *Cellular*) → KE1531 (Decrease, Ca^{2+} binding to Troponin C, *Cellular*) → KE1532 (Decrease, Cardiac contractility, *Organ*) → KE1533 (Decrease, Cardiac ejection fraction, *Organ*) → KE1535 (Heart failure)

A parallel branch runs through KER1760–1762: decreased Ca^{2+} currents → Disruption of intracellular Ca^{2+} mobilization (KE1536) → Disruption of sarcomere assembly (KE1537) → back to decreased contractility.

Without the AOP, a positive result from a Ca^{2+} channel binding assay is ambiguous — Ca^{2+} channel blockers are also antihypertensives. The AOP provides the *dose-response and temporal context*: if Ca^{2+} binding to Troponin C is measurably reduced (KE1531), sarcomere organisation disrupted (KE1537), and contractility falls (KE1532), you now have biologically plausible evidence of a harm trajectory, not just target engagement.

2. AOP480 — Mitochondrial Complex Inhibition → Heart Failure via Decreased ATP

Chain: KE105 (Inhibition, ETC Complexes, *Molecular*) → KE177 (Mitochondrial dysfunction, *Cellular*) → KE40 (Decrease, Mitochondrial ATP production, *Cellular*) → KE1532 (Decrease, Cardiac contractility, *Organ*) → KE2215 (Decrease LV function)

This pathway is highly relevant to anthracycline cardiotoxicity and environmental pollutant exposure. Critically, KE177 (mitochondrial dysfunction) has **7 validated assays in the wiki**: cellular O₂ consumption, $\Delta\psi_m$ (mitochondrial membrane potential), ETS enzymatic activity, ATP content, mitochondrial permeability transition pore (PTP) opening, mtDNA damage, and ROS/oxidative stress measurement. KE105 (ETC complex inhibition) has 5 assays: MTT, cell respiration, luciferin-luciferase ATP determination, cytochrome c domain assays, enzyme activity.

This is a direct example of AOP-driven assay selection: you would deploy KE105 assays first (e.g. Seahorse XF, luciferin-luciferase ATP), then KE177 assays ($\Delta\psi_m$ by JC-1 or TMRM, PTP opening), then contractility readouts in hiPSC-CMs or EHTs. Without the AOP framework, labs routinely run these assays in isolation with no mechanistic connection to each other.

3. AOP555 — hERG (ERG Channel) Inhibition → Heart Failure via QT Prolongation and Early After-Depolarisations

Chain: KE593 (hERG inhibition, *Molecular*) → KE1961 (Prolongation of Action Potential Duration, *Organ*) → KE1962 (QT interval prolongation, *Organ*) → KE2283 (Early after-depolarisations, *Cellular*) → KE1535 (Heart failure)

KE593 has three registered assays: **patch clamp electrophysiology**, **hERG fluorescence polarization**, and the **ToxCast NVS_IC_hKhERGCh assay** — the very assays most commonly run as standalone screens. The AOP makes explicit that hERG inhibition alone is not sufficient: it must propagate to action potential prolongation, then QT prolongation, then triggered activity, before heart failure risk is established. This maps directly onto the ICH E14/S7B paradigm and the Comprehensive in vitro Proarrhythmia Assay (CiPA) initiative, where the AOP logic was implicitly being operationalised without the formal AOP language.

4. AOP552 — L-Type Calcium Channel Blockade → Heart Failure via QT Prolongation and Torsades de Pointes

This is a mechanistically distinct route from AOP261 using the same MIE (KE1529, LTCC blockade). The chain goes through altered action potential (KE698) → QT prolongation (KE1962) → Torsades de Pointes (KE1963) → uncoordinated cardiac contraction (KE2281, *Tissue*) → Heart failure. This AOP illustrates an important AOP network concept: the same MIE (LTCC blockade) can drive adverse outcomes through two divergent pathways — contractility loss (AOP261) and arrhythmia (AOP552). Without the network view, a compound that blocks LTCCs might be assessed against only one pathway, missing the other entirely.

Where AOPs Add Value Beyond Standalone NAMs: Contexts of Use

1. Assay Battery Design and Prioritisation

When you have mechanistic NAMs but no AOP, the choice of which assays to run, and in which sequence, is guided by convention or compound class rather than mechanism. AOP261 and AOP480 make explicit that contractility readouts (KE1532) sit *downstream* of both Ca^{2+} signalling and mitochondrial pathways — so a primary contractility hit should trigger upstream pathway disambiguation (Ca^{2+} vs. mitochondrial) rather than being reported as a standalone negative-or-positive. The KE hierarchy provides a biologically justified decision tree.

2. Weight-of-Evidence Integration

A standalone assay positive is binary: hit or no hit, with the implicit assumption that any hit is equally concerning. An AOP allows semi-quantitative weight-of-evidence: how many KEs in the chain are perturbed? At what concentration relative to dose-response concordance? Are upstream KEs perturbed without downstream propagation (suggesting compensation or threshold effects)? This moves the assessment from signal detection to mechanistic plausibility scoring — precisely what regulators need.

3. Bridging Signal to Regulatory Decision

This is perhaps the most strategically important use. The challenge of NAM-to-regulatory transition is that a positive result in an *in vitro* assay does not constitute evidence of adverse outcome under realistic exposure conditions. AOPs provide the mechanistic justification for *why* a cellular perturbation is biologically plausible to lead to organ-level harm, establishing the *biological plausibility* pillar of a causal inference framework. For cardiac safety, the AOP network anchors data from hiPSC-CM assays, organotypic models, and organ-on-chip platforms to the same adverse outcomes (heart failure, arrhythmia, LV dysfunction) that appear in clinical labels and ICH guidelines.

The OECD's guidance on using AOPs for regulatory purposes specifically supports this use: an AOP does not need to be 100% complete to be regulatory relevant — partial AOP evidence supporting key transitions, especially from MIE through early KEs, is sufficient to justify concern or to support a read-across argument.

4. Identification of Mechanistic Redundancy and Convergence

The cardiac AOP network reveals that heart failure (KE1535) is reached via at least four mechanistically distinct routes — LTCC blockade, hERG inhibition, ETC complex inhibition, and Na/K-ATPase disruption. A compound that hits multiple MIEs simultaneously (as many environmental chemicals and pharmaceuticals do) could be severely underestimated by pathway-specific NAMs. The AOP network perspective enables mixture assessment and multi-mechanism hazard identification.

5. Anchoring New NAMs During Development

If a new NAM is developed — for instance a multielectrode array (MEA) beating frequency assay in hiPSC-CMs — the AOP framework tells you precisely where in the causal chain it sits (likely KE1532 or KE1962 level), which upstream KEs it cannot detect, and which downstream KEs remain unevaluated. This prevents the common error of over-interpreting a new assay's scope.

Transition from Signal Detection to Regulatory Decision-Making

The gap between a NAM signal and a regulatory decision has three components: biological plausibility, human relevance, and confidence calibration. AOPs address all three:

Biological plausibility is established by the KER evidence — each KER in the AOP-Wiki is evaluated for empirical support, biological plausibility, essentiality, and dose-response concordance. A KER like KER1756 (decreased Ca^{2+} currents → decreased Ca^{2+} -TroponinC binding) carries mechanistic weight from decades of cardiac physiology.

Human relevance is demonstrated where AOP KEs are anchored to human biology — KE593 (hERG), KE1529 (LTCC), KE177 (mitochondrial dysfunction) are all well-validated in human cardiac tissue and hiPSC-CMs.

Confidence calibration is what AOPs add over standalone assays. Rather than a pass/fail signal, the weight of evidence across multiple KEs in a chain allows a graduated confidence statement: "compounds A and B both inhibit hERG (KE593), but only A also prolongs action potential duration (KE1961) at the same concentration, and only A shifts QT in the MEA assay (KE1962) — therefore A carries higher arrhythmia liability."

This is the language regulators can act on: not "this compound was positive in a hERG assay" but "this compound activates the hERG → QT prolongation → TdP arm of AOP555 across three independent assays at concentrations consistent with predicted human C_{max} ."

Which Cardiac NAMs and Endpoints Benefit Most from AOP Anchoring

Highest priority — already assay-rich but context-poor:

- **hERG patch clamp / fluorescence polarisation / ToxCast assays (KE593):** currently run as standalone screens. AOP555 anchoring places them as MIE detection for a pathway requiring QT and EAD confirmation before risk conclusion.
- **hiPSC-CM contractility / beating amplitude** (maps to KE1532): commonly reported without mechanistic upstream context. AOP261 and AOP480 tell you this readout could reflect LTCC blockade, ETC inhibition, or sarcomere disruption — upstream disambiguation is essential for mechanistic attribution.

- **MEA/field potential duration** (maps to KE1961/1962): the standard CiPA NAM. AOP552/555 anchor it formally in arrhythmia AOP chains, providing the biological justification for regulatory weight.
- **Mitochondrial membrane potential ($\Delta\psi_m$) and OCR** (KE177): well-validated assays (JC-1, TMRM, Seahorse XF) with 7 registered assay types under KE177. AOP480 anchors these directly to cardiac contractility and LV function — a connection not routinely stated in toxicology study reports.

Endpoints that would most benefit:

- **Ejection fraction / cardiac output** (KE1533, KE2215): currently only accessible in animal models or cardiac imaging. AOPs establish these as the legitimate *in vivo* anchors for *in vitro* upstream signals, supporting a mechanistic bridge argument.
- **Action potential morphology** (KE698): the AOP network distinguishes compounds that alter AP shape via LTCC blockade (AOP552) from those that do so via Na⁺ channel effects (AOP553) — a distinction invisible to MEA beat rate alone.
- **QT interval** (KE1962): shared KE between AOP552 and AOP555, meaning it sits in an AOP network node and carries weight for multiple upstream mechanisms — making it a particularly high-value bridging biomarker.

Summary

The value of AOPs in cardiovascular research is not that they replace mechanistic NAMs — it is that they provide the **causal scaffolding** that individual NAMs cannot supply alone. A positive hERG result means something different when it is embedded in AOP555 (with downstream QT and EAD evidence) than when it stands alone. An mitochondrial toxicity signal means something different when it is connected via AOP480 to ATP deficit, contractility loss, and LV dysfunction than when it is reported as a $\Delta\psi_m$ change in a well plate. The AOP framework transforms isolated mechanistic signals into an integrated, biologically coherent narrative that is both interpretable by non-specialists and actionable by regulators — and that is its core added value.